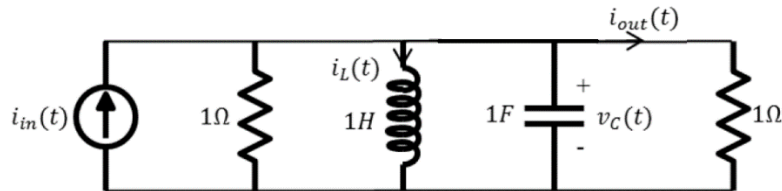


Set A

Q1 [CO1] Consider the circuit shown below, where the excitation is $i_{in}(t) = 2 \sin(30t - 50^\circ) A$.

- Find Thevenin's equivalent circuit seen from the terminals of the rightmost 1Ω resistor.
- Using Thevenin's equivalent circuit, find the time-domain current $i_{out}(t)$ and voltage $v_C(t)$.



Set B

Q1 [CO1] Consider the circuit shown below, where the excitation is $v_{in}(t) = 12 \sin(1t - 60^\circ) V$.

- Find Thevenin's equivalent circuit seen from the terminals of the capacitor.
- Using Thevenin's equivalent circuit, find the time-domain current $i_C(t)$ and voltage $v_C(t)$.

